

## ***Effects of Hibiscus sabdariffa extract powder and preventive treatment (diet) on the lipid profiles of patients with metabolic syndrome (MeSy).***

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Insulin resistance, obesity, hypertension, and dyslipidemia are strongly associated with metabolic syndrome (MeSy), which is considered to be a reversible clinical stage before its evolution to coronary heart disease and diabetes. Currently, the antihypertensive and hypolipidemic properties of aqueous Hibiscus sabdariffa extracts (HSE) have been demonstrated in clinical trials and in vivo experiments. The aim of the present study was to evaluate the effects of a Hibiscus sabdariffa extract powder (HSEP) and a recognized preventive treatment (diet) on the lipid profiles of individuals with and without MeSy according to the National Cholesterol Education Program Adult Treatment Panel III (NCEP-ATP III) criteria. The protocol was a follow-up study carried out in a factorial, randomized design (T1=preventive treatment comprises Diet, T2=HSEP, T3=HSEP+preventive treatment (Diet) X MeSy, non-MeSy individuals). A total daily dose of 100 mg HSEP was orally administered in capsules for one month. The preventive treatment (diet) was selected according to NCEP-ATP III recommendations and adjusted individually. Total cholesterol, LDL-c, HDL-c, VLDL-c, triglycerides, glucose, urea, creatinine, AST, and ALT levels in the blood were determined in all individuals pre- and post-treatment. The MeSy patients treated with HSEP had significantly reduced glucose and total cholesterol levels, increased HDL-c levels, and an improved TAG/HDL-c ratio, a marker of insulin resistance (t-test  $p < 0.05$ ). Additionally, a triglyceride-lowering effect was observed in MeSy patients treated with HSEP plus diet, and in individuals without MeSy treated with HSEP. Significant differences in total cholesterol, HDL-c, and the TAG/HDL-c ratio were found when the means of absolute differences among treatments were compared (ANOVA  $p < 0.02$ ). Therefore, in addition to the well documented hypotensive effects of Hibiscus sabdariffa, we suggest the use of HSEP in individuals with dyslipidemia associated with MeSy.